2020 Q1

The slope field for a first order differential equation is shown. **1**

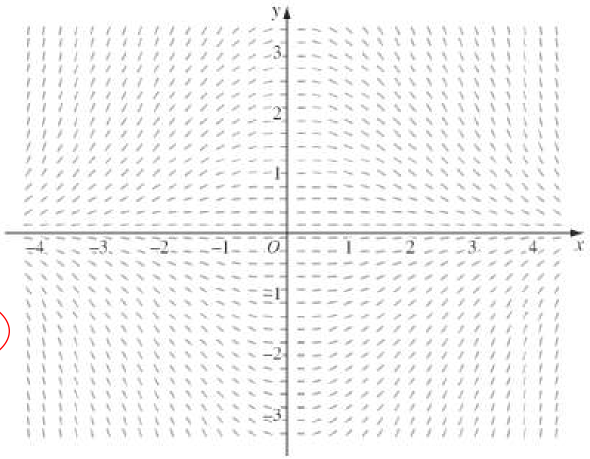
Which of the following could be the differential equation represented?

A. $\dfrac{dy}{dx}=\dfrac{x}{3y}$

B. $\dfrac{dy}{dx}=-\dfrac{x}{3y}$

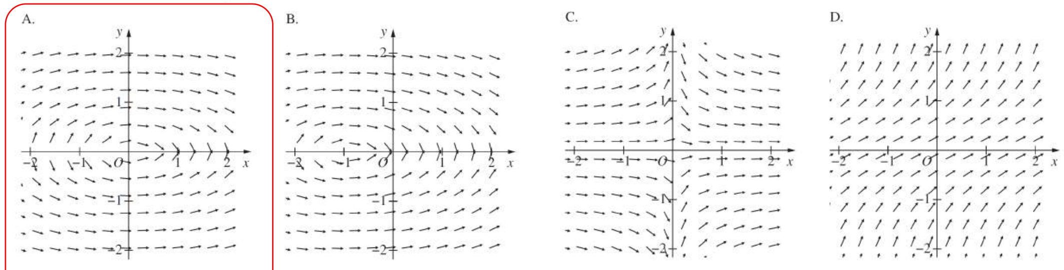
C. $\dfrac{dy}{dx}=\dfrac{xy}{3}$

D. $\dfrac{dy}{dx}=-\dfrac{xy}{3}$



2020 Q2

Which of the following best represents the direction field for the differential equation $\dfrac{dy}{dx}=-\dfrac{x}{4y}$? **1**



2020 Q3

Solve $\dfrac{dy}{dx}=e^{2y}$, finding x as a function of y. **2**

$$\frac{dy}{dx}=e^{2y}$$

$$\frac{dx}{dy}=e^{-2y}$$

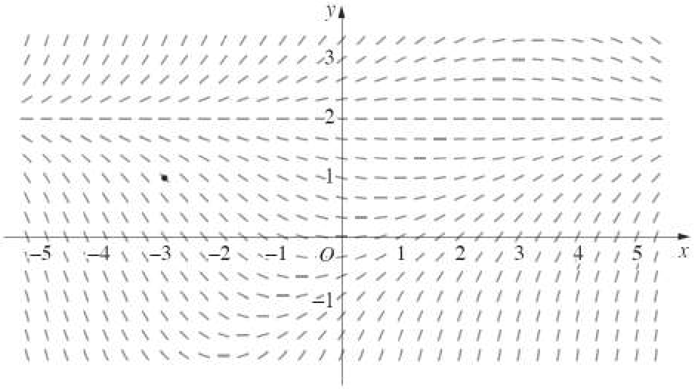
$$x=\int e^{-2y}\,dy=-\frac{1}{2}e^{-2y}+C$$

2020 Q 4

The trajectories of particles in a fluid are described by the differential equation **2**

$$\frac{dy}{dx}=\frac{1}{4}(y-2)(y-x).$$

The slope field for the differential equation is sketched below.

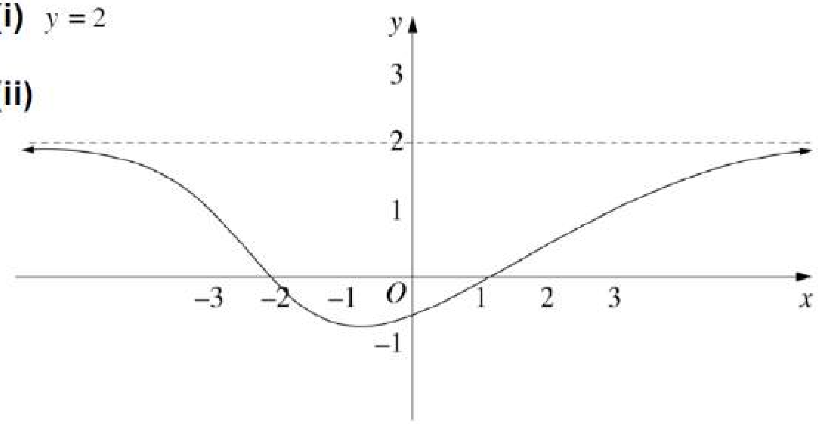


(i) Identify any solutions of the form $y=k$, where k is a constant. **1**

(ii) Draw a sketch of the trajectory of a particle in the fluid which passes through the point $(-3,1)$ and describe the trajectory as $x\to\pm\infty$. **3**

(i) $y=2$

(ii)



The y-coordinate of the particle approaches 2 from below as $x\to\pm\infty$.

2020 Q5

Find the curve which satisfies the differential equation $\frac{dy}{dx} = -\frac{x}{y}$ and passes through the point $(1, 0)$. **3**

$$\frac{dy}{dx} = \frac{-x}{y} \qquad \text{Separating variables, } \int y\,dy = \int -x\,dx$$

$$\therefore \quad \frac{y^2}{2} = \frac{-x^2}{2} + c$$

$$x^2 + y^2 = d, \text{ where } d = 2c$$

The curve passes through $(1, 0)$

So $d = 1^2 + 0^2 = 1$

Hence the equation of D is

$$x^2 + y^2 = 1 \qquad \text{(unit circle)}$$

2021 Q6

Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$. **1**

Which of the following equations best represents this relationship between x and y?

A. $y^2 = x^2 + c$ B. $y^2 = \frac{x^2}{2} + c$ C. $y = x\ln|y| + c$ D. $y = \frac{x^2}{2}\ln|y| + c$

2021 Q7

A bottle of water, with temperature 5°C, is placed on a table in a room. The temperature of the room remains constant at 25°C. After t minutes, the temperature of the water, in degrees Celsius, is T. The temperature of the water can be modelled using the differential equation

$\frac{dT}{dt} = k(T - 25)$ (Do NOT prove this.) where k is the growth constant.

(i) After 8 minutes, the temperature of the water is 10°C. By solving the differential equation, find the value of t when the temperature of the water reaches 20°C. Give your answer to the nearest minute. **3**

(ii) Sketch the graph of T as a function of t. **1**

Sample answer:

$$\int \frac{dT}{T-25} = \int k\,dt$$

$$\therefore kt = \ln(T-25) + c$$

$$\therefore T - 25 = Ae^{kt}$$

when $t = 0$, $T = 5$

$$\therefore -20 = A$$

$$\therefore T = 25 - 20e^{kt}$$

when $t = 8$, $T = 10$

$$\therefore 10 = 25 - 20e^{8k}$$

$$-15 = -20e^{8k}$$

$$e^{8k} = \frac{3}{4}$$

$$8k = \ln\left(\frac{3}{4}\right)$$

$$k = \frac{1}{8}\ln\left(\frac{3}{4}\right)$$

when $T = 20$

$$20 = 25 - 20e^{\frac{1}{8}\ln\left(\frac{3}{4}\right)t}$$

$$-5 = -20e^{\frac{1}{8}\ln\left(\frac{3}{4}\right)t}$$

$$\frac{1}{4} = e^{\frac{1}{8}\ln\left(\frac{3}{4}\right)t}$$

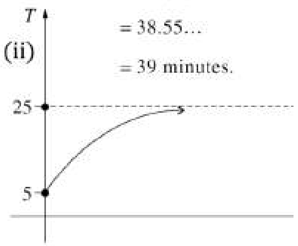
$$\frac{1}{8}\ln\left(\frac{3}{4}\right)t = \ln\left(\frac{1}{4}\right)$$

$$\therefore t = \frac{8\ln\left(\frac{1}{4}\right)}{\ln\left(\frac{3}{4}\right)}$$

$$= 38.55\ldots$$

$$= 39 \text{ minutes.}$$

(ii)



2022 Q 8

Find the particular solution to the differential equation $(x-2)\frac{dy}{dx} = xy$ that passes through the point $(0, 1)$. **4**

Sample answer:

$(x-2)\frac{dy}{dx} = xy$ and $(0, 1)$

$$\int \frac{1}{y}\,dy = \int \frac{x}{x-2}\,dx$$

$$\ln|y| = \int 1 + \frac{2}{x-2}\,dx$$

$$= x + 2\ln|x-2| + C$$

$$= \ln|x-2|^2 + x + C$$

$$\ln\left(\frac{|y|}{(x-2)^2}\right) = x + C$$

$$\frac{|y|}{(x-2)^2} = e^{x+C}$$

$$|y| = e^{x+C}(x-2)^2$$

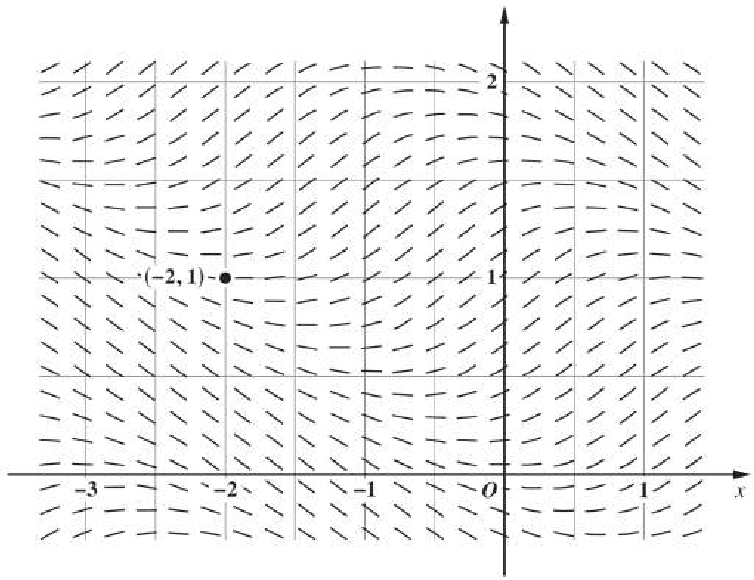
$$y = ke^x(x-2)^2 \qquad \text{where } k = \pm e^C$$

When $x = 0$, $y = 1$ $\quad \therefore \; k = \frac{1}{4}$

$$y = \frac{1}{4}e^x(x-2)^2$$

2023 Q 9

The diagram shows the direction field of a differential equation. A particular solution to the differential equation passes through $(-2, 1)$. **1**



Where does the solution that passes through $(-2, 1)$ cross the y-axis?

A. $y = 1.12$ B. $y = 1.34$ C. $y = 1.56$ D. $y = 1.78$

2024 Q10 Solve the differential equation $\frac{dy}{dx} = xy$, given $y > 0$. Express your answer in the form $y = e^{f(x)}$. **2**

$$\frac{dy}{dx} = xy \qquad \log_e y = \frac{x^2}{2} + C \qquad \text{as } y > 0$$

$$\int \frac{dy}{y} = \int x\,dx \qquad y = e^{\frac{x^2}{2}+C}$$

Answers could include:

$$Ae^{\frac{x^2}{2}}$$

2024 Q11 Find the domain and range of the function that is the solution to the differential equation $\frac{dy}{dx} = e^{x+y}$ and whose graph passes through the origin. **4**

Sample answer:

$$\frac{dy}{dx} = e^{x+y} = e^x e^y$$

$$\int e^{-y}\,dy = \int e^x\,dx$$

$$-e^{-y} = e^x + k$$

$$y = 0,\ x = 0$$

$$-1 = 1 + k$$

$$k = -2$$

$$-e^{-y} = e^x - 2$$

$$e^{-y} = 2 - e^x$$

$$-y = \ln(2 - e^x)$$

$$y = -\ln(2 - e^x)$$

$$e^x < 2$$

$$x < \ln 2$$

Domain is $(-\infty, \ln 2)$

To find the range:

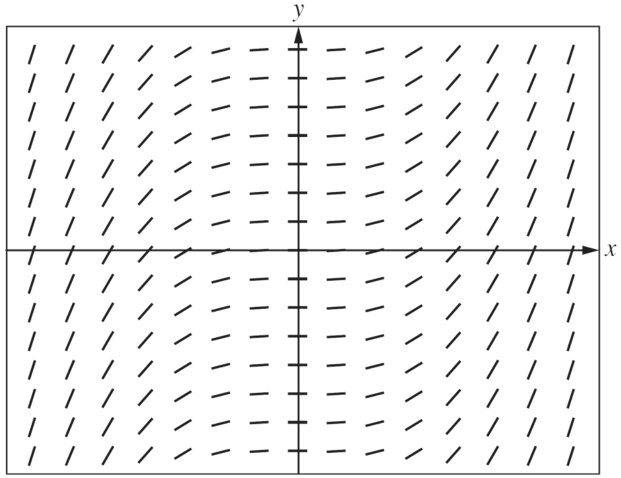
$$x \to -\infty \qquad e^x \to 0 \qquad y \to -\ln 2$$

$$x \to \ln 2 \qquad e^x \to 2 \qquad \ln(2 - e^x) \to -\infty \qquad y \to \infty$$

and $\frac{dy}{dx} = e^{x+y} > 0$ so graph is increasing

Range is $(-\ln 2, \infty)$

2025 Q12 A slope field is shown. **1**



Which of the following could be the differential equation represented by the slope field?

A. $\boxed{\frac{dy}{dx} = x^2}$

B. $\frac{dy}{dx} = x^2 + C,\ C \neq 0$

C. $\frac{dy}{dx} = x^3$

D. $\frac{dy}{dx} = x^3 + C,\ C \neq 0$

2020 Q1

The population of foxes on an island is modelled by the logistic equation $\frac{dy}{dt} = y(1-y)$,
where y is the fraction of the island's carrying capacity reached after t years.
At time $t = 0$, the population of foxes is estimated to be one-quarter of the island's carrying capacity.

(i) Use the substitution $y = \frac{1}{1-w}$ to transform the logistic equation to $\frac{dw}{dt} = -w$. **2**

(ii) Using the solution of $\frac{dw}{dt} = -w$, find the solution of the logistic equation for y satisfying the initial conditions. **2**

(iii) How long will it take for the fox population to reach three-quarters of the island's carrying capacity? **2**

(i) $\frac{dy}{dt} = \frac{w'}{(1-w)^2}$ and $y(1-y) = \frac{-w}{(1-w)^2}$, so $\frac{dw}{dt} = -w$.

(ii) $w = -3$ when $t = 0$ so $w = -3e^{-t}$. Thus $y = \frac{1}{1+3e^{-t}}$.

(iii) $1 + 3e^{-t} = \frac{4}{3}$ so $e^{-t} = \frac{1}{9}$ so $t = \ln 9$ years.

2021 Q 2

In a certain country, the population of deer was estimated in 1980 to be 150 000.

The population growth is given by the logistic equation $\frac{dP}{dt} = 0.1P\left(\frac{C-P}{C}\right)$ **4**

where t is the number of years after 1980 and C is the carrying capacity.

In the year 2000, the population of deer was estimated to be 600 000.

Use the fact that $\frac{C}{P(C-P)} = \frac{1}{P} + \frac{1}{C-P}$ to show that the carrying capacity is approximately 1 130 000.

Sample answer:

$$\frac{dP}{dt} = 0.1P\left(\frac{C-P}{C}\right)$$

$$\int \frac{C}{P(C-P)}\,dP = \int 0.1\,dt$$

$$\int \left(\frac{1}{P} + \frac{1}{C-P}\right)dP = \int 0.1\,dt$$

$\ln|P| - \ln|C-P| = 0.1t + k$ where k is a constant

$$\ln\left|\frac{P}{C-P}\right| = 0.1t + k$$

$\frac{P}{C-P} = Ae^{0.1t}$ where $A = e^k$ is a constant

When $t = 0$, $P = 150\,000$ so $\frac{150\,000}{C - 150\,000} = A$ ①

When $t = 20$, $P = 600\,000$ so $\frac{600\,000}{C - 600\,000} = Ae^2$ ②

Substituting ① into ②: $\frac{600\,000}{C-600\,000} = \frac{150\,000}{C-150\,000}e^2$

Taking the reciprocal of both sides

$$\frac{C-600\,000}{600\,000} = \frac{C-150\,000}{150\,000}e^{-2}$$

$$150\,000(C-600\,000) = 600\,000(C-150\,000)e^{-2}$$

$$C\left(150\,000 - 600\,000e^{-2}\right) = 150\,000 \times 600\,000\left(1-e^{-2}\right)$$

$$C = \frac{150\,000 \times 600\,000\left(1-e^{-2}\right)}{150\,000 - 600\,000e^{-2}} \simeq 1\,131\,121$$

$$\simeq 1\,131\,000$$

2022 Q3

In a room with temperature 12°C, coffee is poured into a cup. The temperature of the coffee when it is poured into the cup is 92°C, and it is far too hot to drink.

The temperature, T, in degrees Celsius, of the coffee, t minutes after it is made, can be modelled using the differential equation $\frac{dT}{dt} = k(T - T_1)$, where k is the constant of proportionality and T_1 is a constant.

(i) It takes 5 minutes for the coffee to cool to a temperature of 76°C. Using separation of variables, solve the given differential equation to show that $T = 12 + 80e^{\frac{t}{5}\ln\left(\frac{4}{5}\right)}$. **3**

(ii) The optimal drinking temperature for a hot beverage is 57°C. Find the value of t when the coffee reaches this temperature, giving your answer to the nearest minute. **1**

$\frac{dT}{dt} = k(T - T_1)$

Note that as $t \to +\infty$, $\frac{dT}{dt} \to 0$ and T approaches room temperature

so T_1 = room temperature = 12.

Also T decreases with time, so $T - T_1 > 0$.

$\frac{dT}{T-12} = k\,dt$ (separate variables)

$\ln(T - 12) = kt + c$ where c is a constant

$T - 12 = Ae^{kt}$ where $A = e^c$

$T = 12 + Ae^{kt}$

When $t = 0$: $92 = 12 + A$ so $A = 80$

When $t = 5$: $76 = 12 + 80e^{5k}$

$e^{5k} = \frac{4}{5}$

$5k = \ln\left(\frac{4}{5}\right)$

$k = \frac{1}{5}\ln\left(\frac{4}{5}\right)$

Finally $T = 12 + 80e^{\frac{1}{5}\ln\left(\frac{4}{5}\right)t}$ for $t \geq 0$.

(ii) For $T = 57$

$57 = 12 + 80e^{\frac{1}{5}\ln\left(\frac{4}{5}\right)t}$

$e^{\frac{1}{5}\ln\left(\frac{4}{5}\right)t} = \frac{57-12}{80} = \frac{9}{16}$

$\frac{1}{5}\ln\left(\frac{4}{5}\right)t = \ln\left(\frac{9}{16}\right)$

$t = \frac{5\ln\left(\frac{9}{16}\right)}{\ln\left(\frac{4}{5}\right)} = 12.89\ldots \approx 13$ minutes

The coffee will reach a temperature of 57°C after about 13 minutes.

2025 Q4

It is given that $\frac{dy}{dx} = \frac{5}{y}$ and $y = -4$ when $x = 0$. **3**

Find y as a function of x.

Since $\frac{dy}{dx} = \frac{5}{y}$,

$\therefore \int y\,dy = \int 5\,dx$

$\frac{y^2}{2} = 5x + C$

When $x = 0$, $y = -4$.

$\therefore \frac{16}{2} = 0 + C$

$C = 8$

ie $\frac{y^2}{2} = 5x + 8$

$y^2 = 10x + 16$

$y = \pm\sqrt{10x + 16}$

When $x = 0$, $y = -4$

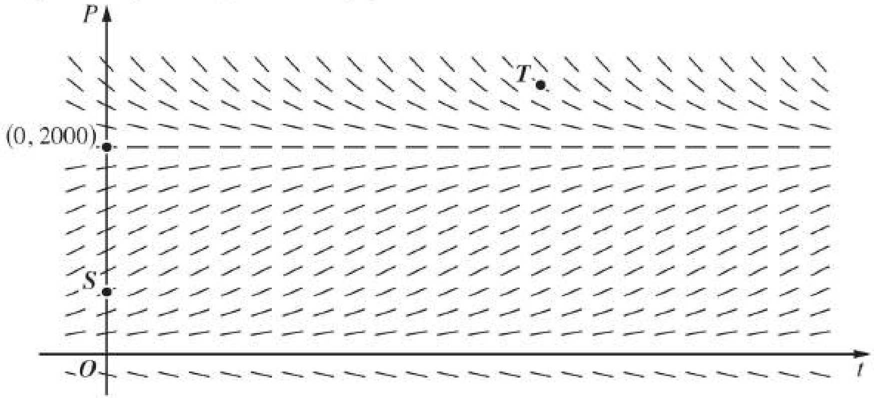
$\therefore y = -\sqrt{10x + 16}$

2024
Q5

In an experiment, the population of insects, $P(t)$, was modelled by the logistic differential equation $\dfrac{dP}{dt} = P(2000 - P)$

where t is the time in days after the beginning of the experiment.

The diagram shows a direction field for this differential equation, with the point S representing the initial population.



(i) Explain why the graph of the solution that passes through the point S cannot also pass through the point T. **1**

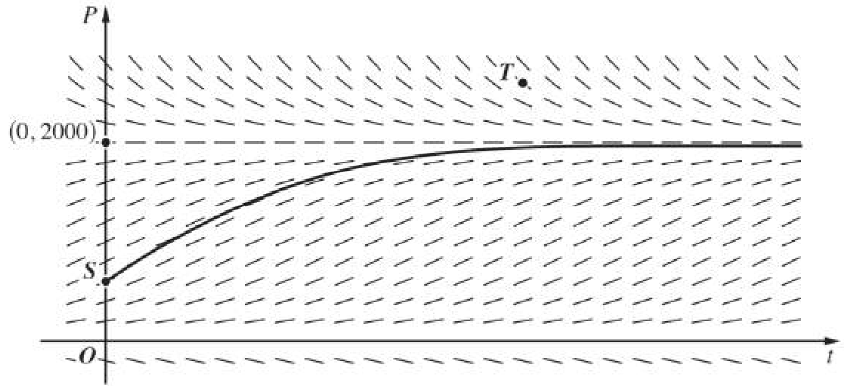
(ii) On the diagram provided on page 1 of the Question 13 Writing Booklet, clearly sketch the graph of the solution that passes through the point S. **1**

(iii) Find the predicted value of the population, $P(t)$, at which the rate of growth of the population is largest. **2**

Question 13 (a) (i)

All graphs passing through T must be decreasing to the left of T. So the y-intercept of such a curve, if any y-intercept, must be above T. The point S isn't above T so S can't be on any such curve.

Question 13 (a) (ii)



Question 13 (a) (iii)

Sample answer:

Rate of growth is a maximum when $\dfrac{dP}{dt}$ is a maximum.

ie when $\dfrac{d^2P}{dt^2} = 0$

$$0 = 1 \times (2000 - P) - P$$
$$2P = 2000$$
$$P = 1000$$

From the direction field it can be seen that the slope at $P = 1000$ is the maximum slope. (Rather than a minimum slope or other type of stationary point.)